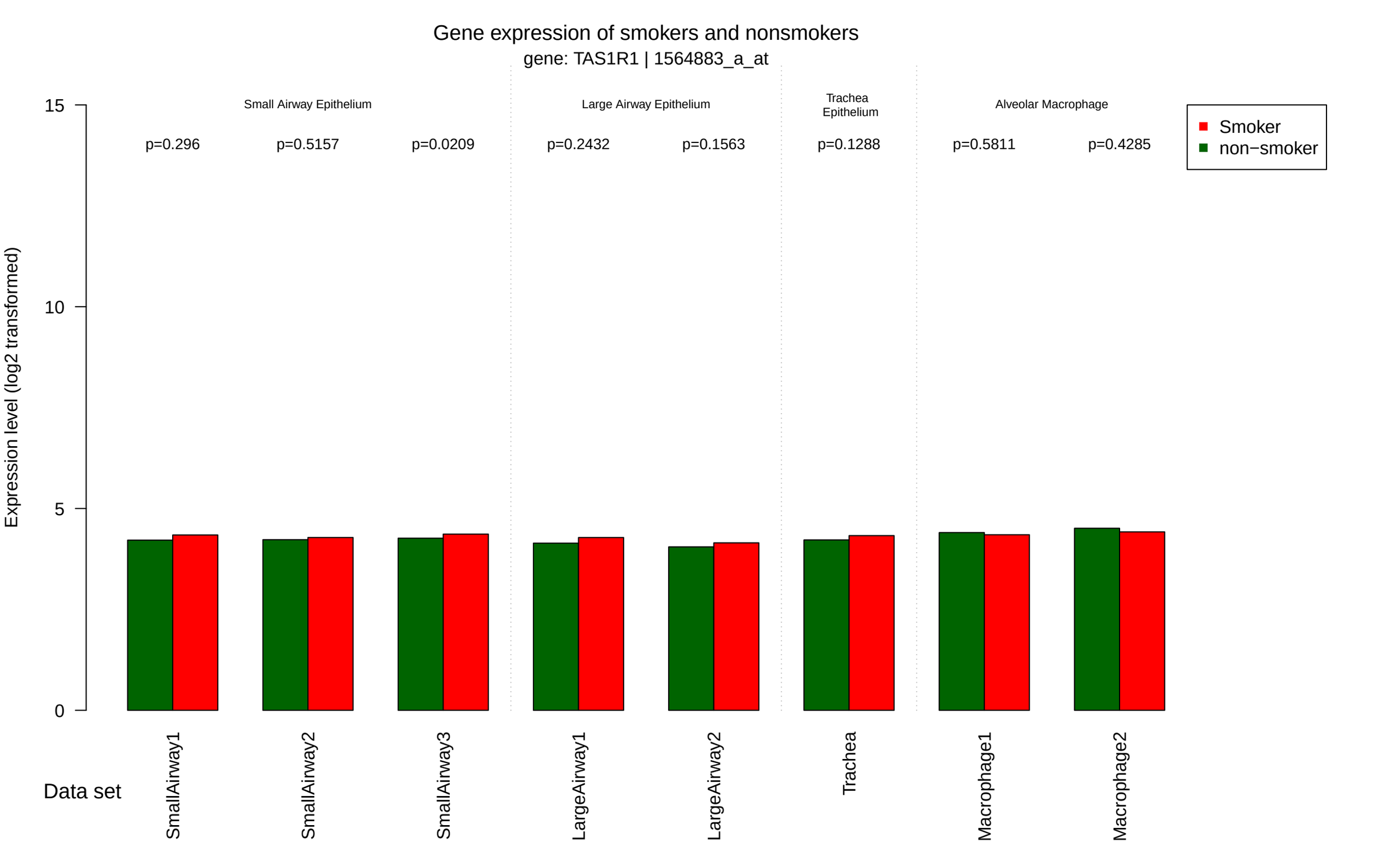
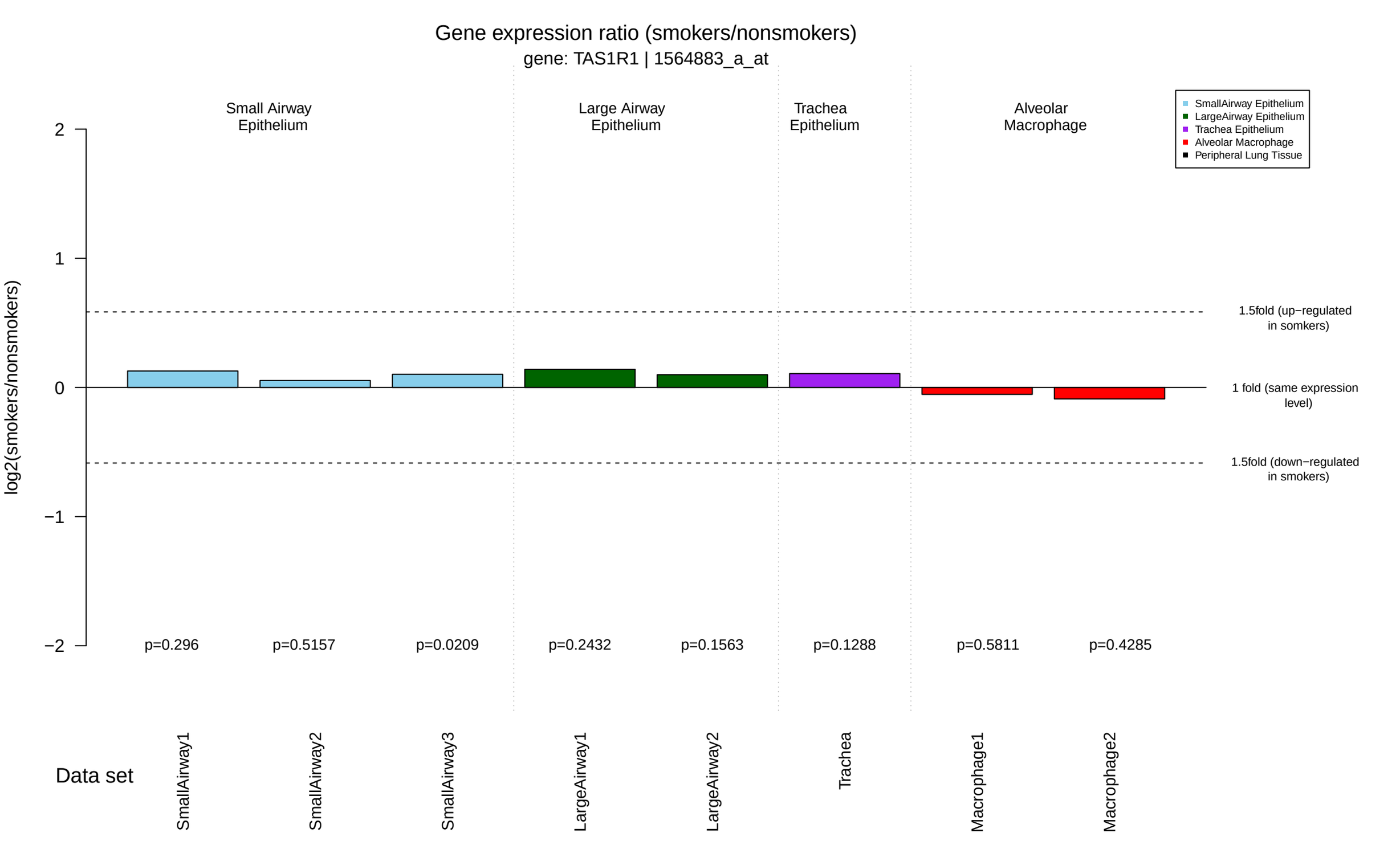
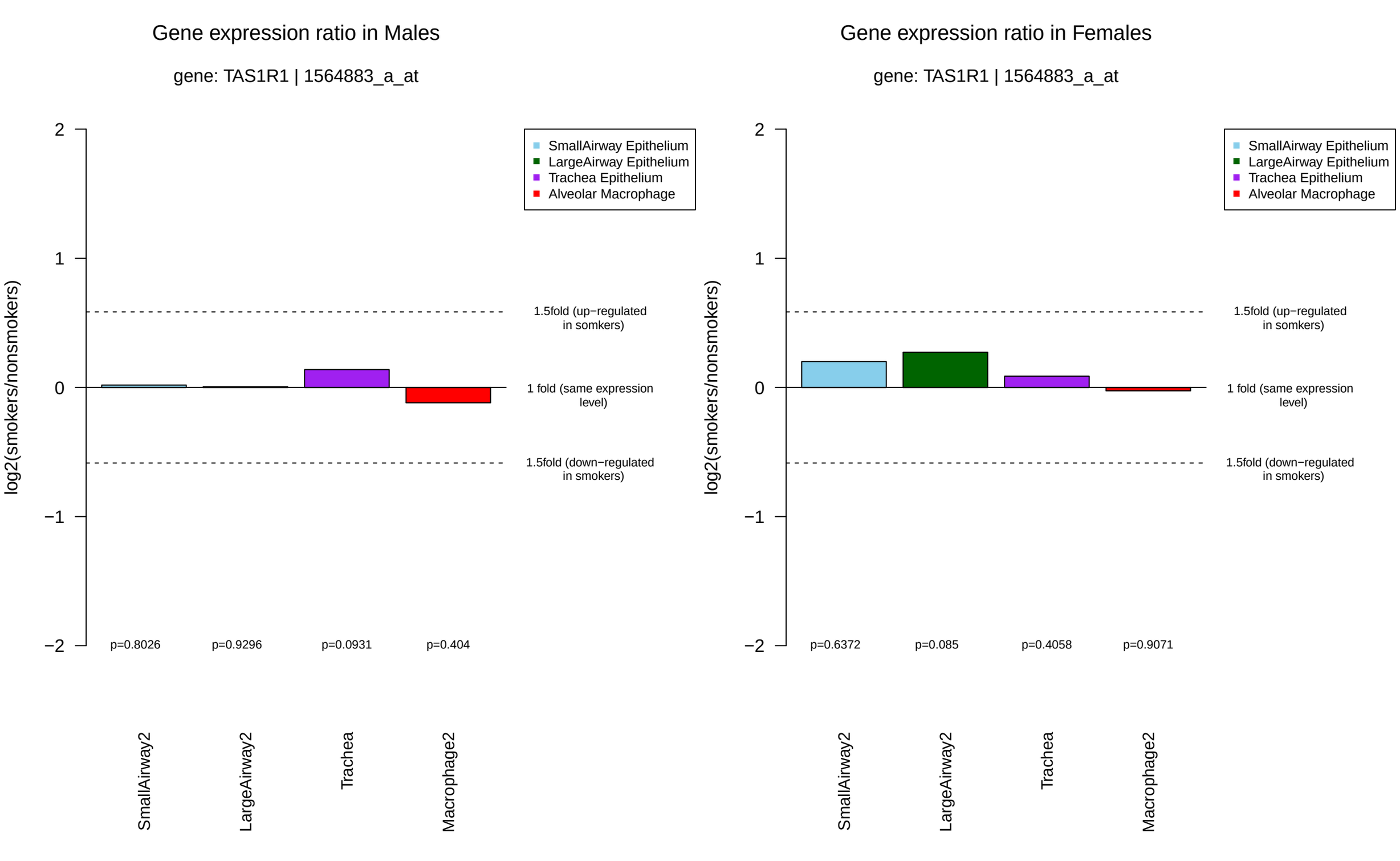
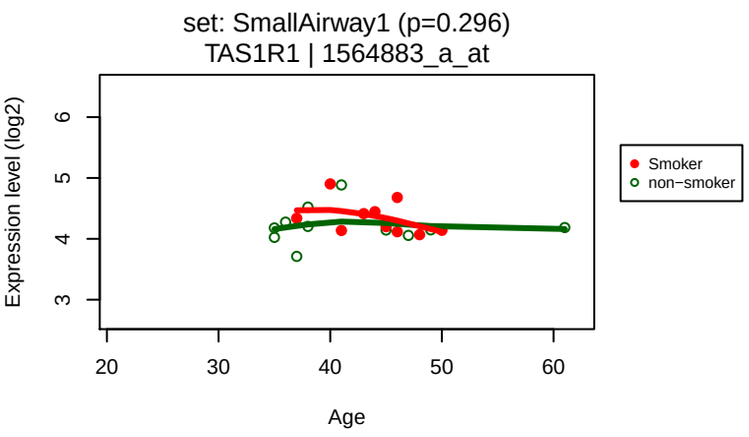
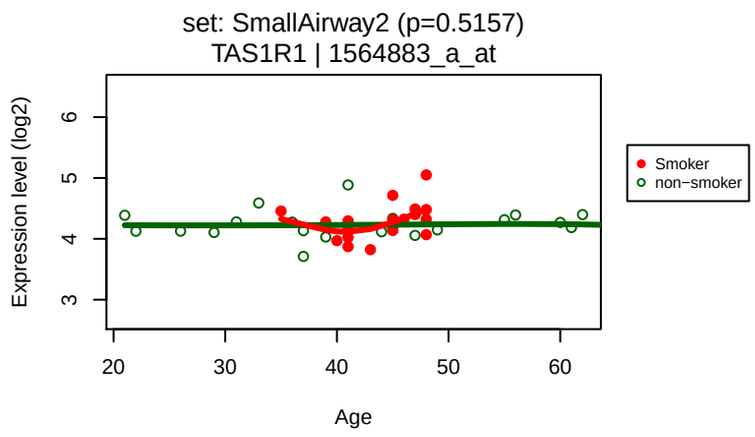
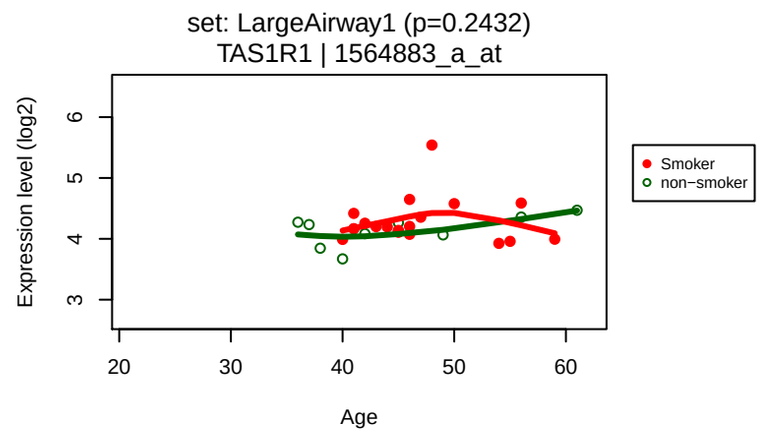

Gene expression of smokers and nonsmokers
gene: TAS1R1 | 1564883_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.296
p=0.5157
p=0.0209
p=0.2432
p=0.1563
p=0.1288
p=0.5811
p=0.4285
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

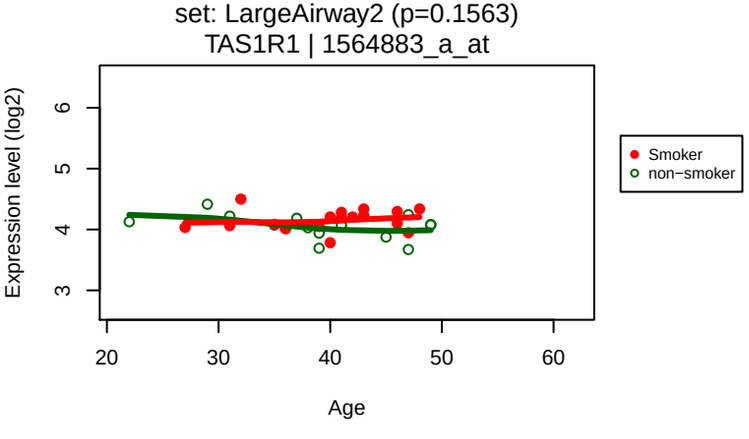

Gene expression ratio (smokers/nonsmokers)
gene: TAS1R1 | 1564883_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.296
p=0.5157
p=0.0209
p=0.2432
p=0.1563
p=0.1288
p=0.5811
p=0.4285
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

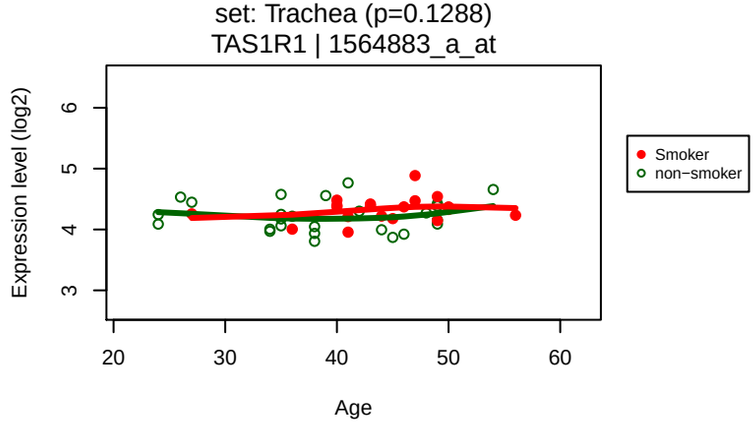

Gene expression ratio in Males
gene: TAS1R1 | 1564883_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8026
p=0.9296
p=0.0931
p=0.404
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS1R1 | 1564883_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6372
p=0.085
p=0.4058
p=0.9071
SmallAirway2
LargeAirway2
Trachea
Macrophage2

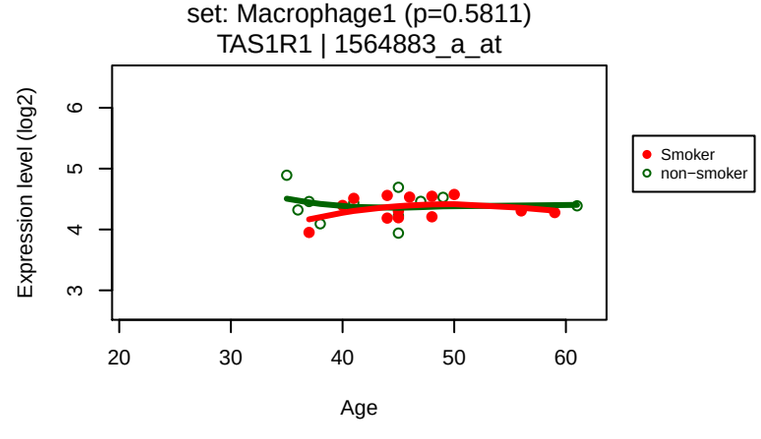

set: SmallAirway1 (p=0.296)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker

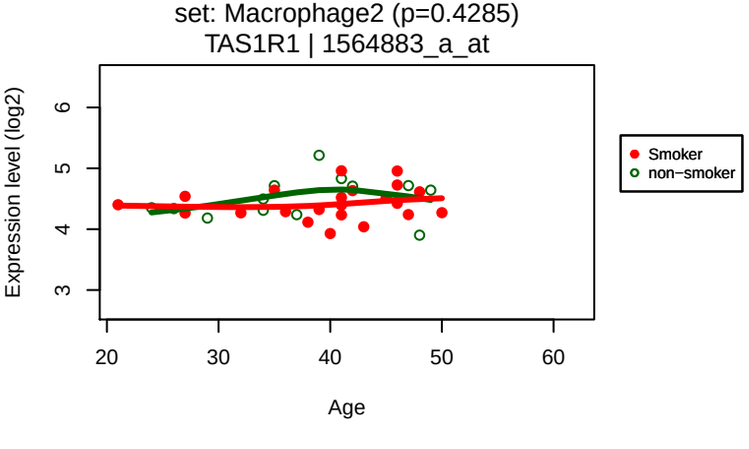

set: SmallAirway2 (p=0.5157)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker

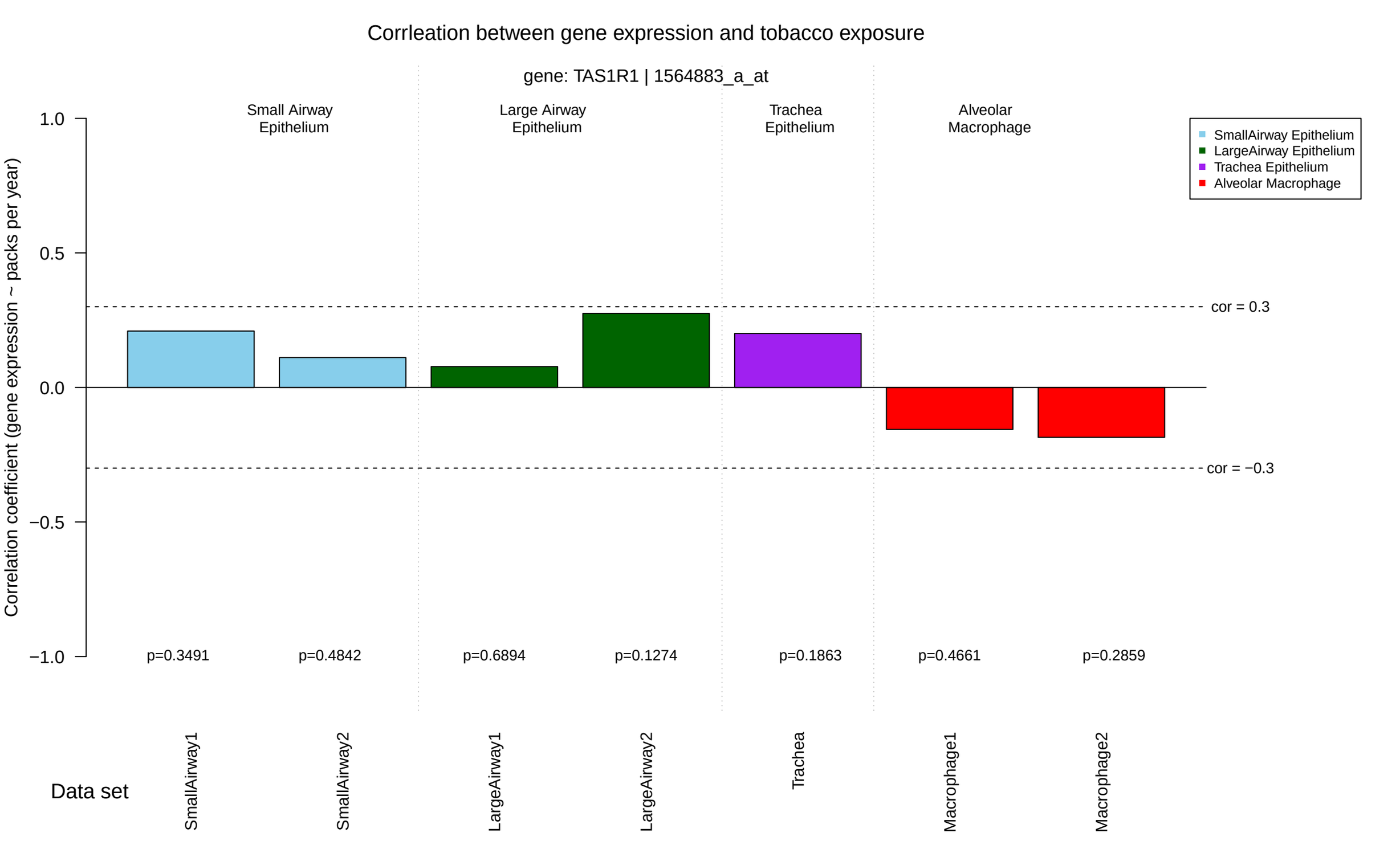

set: LargeAirway1 (p=0.2432)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1563)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1288)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5811)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4285)
TAS1R1 | 1564883_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS1R1 | 1564883_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3491
p=0.4842
p=0.6894
p=0.1274
p=0.1863
p=0.4661
p=0.2859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2